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ORIGINAL ARTICLE

Comparing the accuracy of two machine learning models in detection and classification of periapical lesions using periapical radiographs

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Abstract

Background Previous deep learning-based studies were mainly conducted on detecting periapical lesions; limited information in classification, such as the periapical index (PAI) scoring system, is available. The study aimed to apply two deep learning models, Faster R-CNN and YOLOv4, in detecting and classifying periapical lesions using the PAI score from periapical radiographs (PR) in three different regions of the dental arch: anterior teeth, premolars, and molars.

Methods Out of 2658 PR selected for the study, 2122 PR were used for training, 268 PR were used for validation and 268 PR were used for testing. The diagnosis made by experienced dentists was used as the reference diagnosis.

Results The Faster R-CNN and YOLOv4 models obtained great sensitivity, specificity, accuracy, and precision for detecting periapical lesions. No clear difference in the performance of both models among these three regions was found. The true prediction of Faster R-CNN was 89%, 83.01% and 91.84% for PAI 3, PAI 4 and PAI 5 lesions, respectively. The corresponding values of YOLOv4 were 68.06%, 50.94%, and 65.31%.

Conclusions Our study demonstrated the potential of YOLOv4 and Faster R-CNN models for detecting and classifying periapical lesions based on the PAI scoring system using periapical radiographs.

Keywords Deep learning · Periapical lesion · Periapical index · CNN

Introduction

Periapical lesions are a common pathological condition encountered in dentistry with an incidence of 23–31% in the population [1]. These lesions are typically associated with infections or inflammatory reactions, characterized by bone destruction in the peri-radicular tissues [2]. Thus, apical periodontitis is frequently diagnosed via radiography, such as periapical radiographs (PR), panoramic radiographs, or cone-beam computed tomography scans (CBCT). A systematic review showed CBCT provided the highest accuracy in the detection of periapical lesions, followed by PR [3]. However, CBCT uses a higher radiation dose which is not justified for routine diagnosis and follow-up [4] and can be easily affected by metal artifacts (metal inlay/onlay or adjacent implants) [5]. PR could sufficiently provide information about root morphology, periodontal ligament, alveolar bone, and periapical radiolucency with a modest dose of radiation. Therefore, they are still commonly used for dental checkups and apical periodontitis follow-ups [4].

In recent years, deep learning-based approaches have become increasingly popular in the field of dentistry as a way to provide support or replace time-consuming processes and minimize errors caused by inexperience [6, 7]. Two of the most widely used deep learning models in object detection and classification are Faster R-CNN and YOLOv4 [8]. Faster R-CNN is a two-stage model that first proposes regions of interest and then classifies the objects within those regions. On the other hand, YOLOv4 is a one-stage model that simultaneously predicts object boundaries and class probabilities [9]. Both models have

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demonstrated robustness to noise and variations in the images, making them suitable for clinical applications. A recent study reported that a deep CNN was comparable to the ability of specialists to detect periodontal bone loss in panoramic radiographs [10].

Previous studies were mainly conducted on detecting periapical lesions, limited information in classification, such as the PAI scoring system, is available rather than one study by Moidu et al. [11]. The periapical index (PAI) score is a widely used method for evaluating the severity of periapical lesions on dental radiographs. The PAI scoring system assigns a score of 1–5 to each tooth, with 1 indicating no visible apical changes and 5 indicating a large apical radiolucency with evidence of a widened periodontal ligament [12]. It could provide a standardized way of quantifying the severity of periapical lesions and allow for better communication among dental professionals about treatment options and prognosis. However, traditional methods of manual diagnosing and classifying periapical lesions by dentists can be subjective and prone to interpretation errors, especially in complex cases.

Anatomical factors depending on regions of the dental arch such as maxillary sinus, incisive and mental foramen, inferior alveolar nerve, shape of the root and the thickness of the surrounding bone might hamper the performance of machine learning models. Therefore, when testing the performance of these models, the anatomical factors should be considered.

In this study, we aimed to apply the performance of 2 state-of-the-art deep learning models, Faster R-CNN and YOLOv4, in diagnosing and classifying periapical lesions using the PAI score from PR with three different regions of the dental arch: anterior teeth, premolars, and molars. We hypothesized that these models could accurately detect and classify periapical lesions in all three regions of the dental arch, but there might be differences in performance depending on the region tested.

Materials and methods

Data set

The study was conducted with approval from the Hanoi Medical University Ethics Committee, number HMUIRB463.

The initial data used for the study were 9728 PR recruited from 7677 patients (60.13% males, 39.87% females; mean ± SD age: 37.31 ± 16.52; range: 16–96 years old). All patients were examined and taken PR using X70 FONA (FONA, Milan, Italy) periapical X-ray machine and PXPIS image plating at the High-quality Dental Treatment Center, School of Dentistry, Hanoi Medical University, from December 2016 to March 2021.

Labelling

9728 PR were assessed to detect and classify periapical lesions based on the PAI scoring system by 3 experienced dentists. All of them had more than 5 years of clinical experience. Three assessors underwent calibration using 20 PR to address discrepancies between them. Subsequently, they were recalibrated using 40 additional PR, resulting in a Kappa index of 0.864.

Three examiners meticulously detected and annotated the periapical lesions based on PAI scores (PAI 3, PAI 4 and PAI 5). PAI 1 and PAI 2 lesions are regarded to be within normal levels and were removed according to previous studies [13–16]. Periapical lesions obtained agreement from at least 2 examiners will be selected with the annotation of the majority. Periapical lesions with 3 different annotations will be eliminated. Finally, 2658 PR were selected for the study (Table 1). Annotation was performed using Roboflow software (Roboflow, Inc., Des Moines, Iowa, USA).

The numbers of periapical lesions for PAI 3, PAI 4, and PAI 5 scores were 1718, 809, and 675, respectively. 2658 PR then was randomly allocated in ratios of 80%, 10% and 10% for training, validation and testing.

Training and validation

The training dataset consisting of 2122 images was formatted in PASCAL VOC for Faster R-CNN and COCO for YOLOv4. All images were resized to a uniform resolution of 640-by-640 pixels to standardize input sizes. Image augmentation techniques such as flipping and rotation were employed to enhance dataset quality. Both models were implemented in Python using the PyTorch backend and trained on a computer equipped with an Intel® Core™ i7 CPU (3.00 GHz), 16 GB RAM, and an 11 GB GPU.

For validation, 268 images were used, and pretraining weights were utilized to expedite training time and address convergence issues. Each suspected periapical lesion was assigned a label code of 3, 4, or 5. The final output label

Table 1 Periapical lesions distribution in the data set

PAI scores	Training	Validation	Test
3	1338	189	191
4	635	68	106
5	581	45	49
Total	2554	302	346

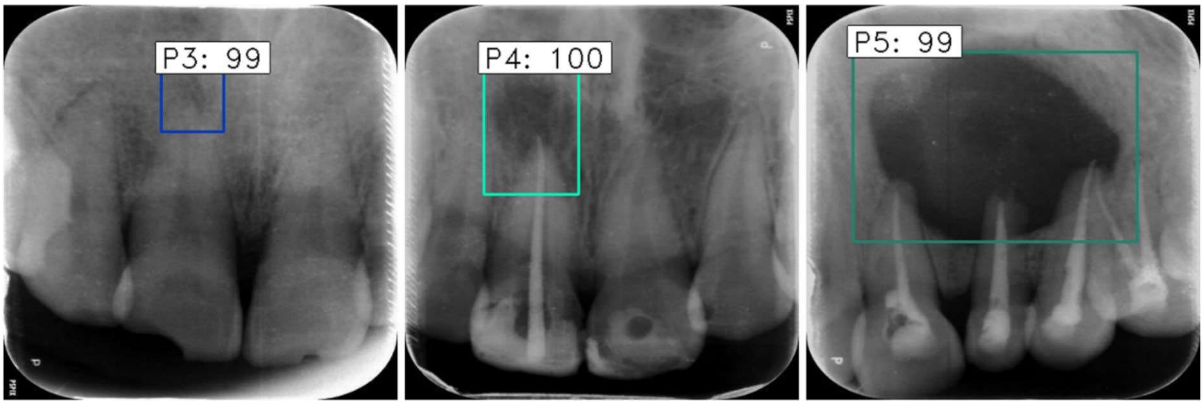

Fig. 1 Apical lesions detected and classified by machine learning models. *P3: PAI3, P4: PAI4, P5: PAI 5*

for each image represented the amalgamation of all diagnoses without specifying the lesion location (Fig. 1).

Deep learning models

Faster R-CNN

In this study, 2 deep learning models were utilized, in which the Faster R-CNN architecture was the first model for classification of Periapical Index (PAI) scores 3, 4, and 5 on periapical radiographs (PR). The Faster R-CNN model consists of a region proposal network (RPN) for generating region proposals and a Region-based CNN (RCNN) for object detection within these proposals.

Model configuration

Backbone network: The backbone network impacts the feature extraction process, and in this study backbone network was designed for better features from input PR images. Considering the complex nature of dental images, ResNet50 was used because it has demonstrated strong feature extraction capabilities in similar tasks.

Region proposal network (RPN): The RPN is responsible for proposing candidate object bounding boxes. It comprises convolutional layers followed by 2 sibling fully connected layers: one for bounding box regression and the other for objectness score prediction. We customize box scales in the RPN to generate proposals specifically targeting regions indicative of PAI scores 3, 4, and 5.

Model training

The balance between model complexity and computational efficiency is permanent while model parameters are explored. Deeper networks may capture finer features relevant to PAI score classification, but we aim to avoid overfitting and excessive computational burden. Therefore, the major options for the Faster-RCNN training process were selected as follows: using a pre-trained model, using batch size 16, learning rate 0.00001, flip and rotation image augmentation.

YOLO v4

The second deep learning model utilized was the YOLOv4 architecture for the classification of periapical index (PAI) scores 3, 4, and 5 on periapical radiographs (PR). YOLOv4 is a state-of-the-art real-time object detection model known for its speed and accuracy.

Model configuration

Backbone network: YOLOv4 typically employs a CSPdarknet-53 backbone network, a lightweight yet powerful CNN architecture. This backbone is adept at extracting features from diverse image domains, including dental radiographs, making it suitable for our task.

Object Detection Head: YOLOv4 utilizes a single-stage object detection approach, where object detection and classification are performed simultaneously. The object detection head comprises convolutional layers responsible for predicting bounding boxes and associated class probabilities. We customize this head to detect regions corresponding to PAI scores 3, 4, and 5.

Model training

The major options of the YOLOv4 training process were selected as follows: using a pre-trained model, using batch size 12, learning rate 0.00001, image augmentation methods were used.

Testing and evaluation

A total of 268 PR which are not included in the training, or the validation set were used for testing. The 2 models were then evaluated based on the results of the testing set.

To evaluate the effects of anatomical factors on deep learning models' diagnosis, the upper and lower tooth arches were divided into 3 separate regions: anterior teeth, premolars, and molars.

Apical lesions were assessed as a dichotomized variable (healthy/ disease) and as PAI scores (healthy, 3, 4, 5). Common parameters were used to evaluate deep learning performance using the diagnosis of experienced dentists as the reference diagnosis. TP: true positives, the number of cases that were correctly classified as positive; FP: false positives, the number of cases that were incorrectly classified as positive; FN: false negatives, the number of instances that were incorrectly classified as negative.

Sensitivity (true-positive rate) measures the proportion of positives that are correctly identified as follows:

$$\text{Sensitivity} = \text{TP}/\ (\text{TP} + \text{FN})$$

Specificity (true-negative rate) measures the proportion of negatives that are correctly identified as follows:

$$\text{Specificity} = \text{TN}/\ (\text{TN} + \text{FP})$$

Accuracy can be represented as the number of classified datasets divided by the total number of data test sets and is defined as follows:

$$\text{Accuracy} = (\text{TP} + \text{TN})\ /\ (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

The precision rate indicates the correct prediction of the number of categories divided by the total number of data falling into that category as follows:

$$\text{Precision} = \text{TP}/\ (\text{TP} + \text{FP}).$$

Statistical analysis

The data were analyzed using SPSS version 22 (IBM, Armonk, NY, USA). Cohen kappa statistic was used to evaluate the inter-reliability of the agreement of the examiners.

Results

Detecting apical lesions

Table 2 shows the assessing parameters of two models when assessing the apical lesion as dichotomization. For the overall arch, the Faster R-CNN model obtained great sensitivity, specificity, accuracy, and precision for detecting periapical lesions at 96.2%, 95.41%, 95.74%, and 93.37%, respectively. Meanwhile, these numbers of YOLOv4 were 79.47%, 91.47%, 87.0%, and 86.75%.

Regarding separate tooth types, no significant difference was observed between maxillary teeth and mandibular teeth or between anterior teeth, premolars, and molars.

Classification based on PAI scoring system

The YOLOv4 showed true prediction in PAI-based classification at 68.06% for PAI 3, 50.94% for PAI 4 and 65.31 for PAI 5 periapical lesions (Table 3).

The true prediction of Faster R-CNN was 89%, 83.01% and 91.84% for PAI 3, PAI 4 and PAI 5 lesions, respectively (Table 4).

Table 2 Performance of two models in detecting periapical lesion

	Model	Sensitivity %	Specificity%	Accuracy %	Precision %
YOLOv4	Maxi-anterior	76.65	94.87	85.23	92.98
	Maxi-Premolars	76.67	92.74	89.61	71.88
	Maxi-Molars	81.73	87.23	84.34	87.63
	Mandi-anterior	80	94.54	89.41	88.89
	Mandi-Premolars	92	94.38	93.86	82.14
	Mandi-Molars	77.90	89.15	83.43	88.16
	Total	**79.47**	**91.47**	**87.0**	**86.75**
Faster-R-CNN	Maxi-anterior	94.36	94.87	94.63	94.37
	Maxi-Premolars	100	96.77	97.4	88.24
	Maxi-Molars	98.08	97.87	97.98	98.08
	Mandi-anterior	93.33	94.54	94.12	90.32
	Mandi-Premolars	100	95.5	96.49	86.21
	Mandi-Molars	94.19	91.57	92.9	92.05
	Total	**96.2**	**95.41**	**95.74**	**93.37**

Table 3 Performance of YOLOv4 in PAI-based classification

PAI scores	True prediction (%)	False prediction (%)			
		Healthy (non-detect)	3	4	5
3	68.06	24.08	–	7.86	0
4	50.94	17.92	31.14	–	0
5	65.31	12.44	12.24	10.21	–

Table 4 Performance of Faster R-CNN in PAI-based classification

PAI scores	True prediction (%)	False prediction (%)			
		Healthy (non-detect)	3	4	5
3	89.00	6.81	–	3.66	0.53
4	83.01	0	14.15	–	2.84
5	91.84	0	2.04	6.12	–

Both models gave the lowest true prediction for PAI 4 compared to other PAI scores, which was often misinterpreted as PAI 3.

Discussion

Our study tested 2 deep learning models, Faster R-CNN and YOLOv4, in detecting and PAI-based classifying periapical lesions using PR. We also tested these two models on three different regions of the dental arch: anterior teeth, premolars, and molars. Our findings provided deep insights into the potential of Faster R-CNN and YOLOv4 on classification of periapical lesions for dental practice, as well as the impact of anatomical factors on their performance.

Detecting periapical lesions

The first goal of our study was to apply 2 deep learning models, Faster R-CNN and YOLOv4, in automatically detecting periapical lesions on PR. The Faster R-CNN demonstrated an excellent level of diagnostic performance compared to the diagnosis of experienced dentists, with overall sensitivity, specificity, accuracy, and precision of 96.20%, 95,41%, 95.74%, and 93.37%, respectively. Meanwhile, the corresponding values in YOLOv4 were 79.47%, 91.47%, 87.0%, and 86.75%. Our study's findings on the high accuracy of deep learning models align with the broader trend observed in recent research. A 2023 systematic review by Soroush Sadr [17] analyzing 18 studies confirmed the effectiveness of deep learning in identifying periapical lesions on dental radiographs. This further underscores the potential of deep learning for automating periapical lesion detection in clinical practice.

Several previous works studying periapical lesions were thoroughly examined. However, it should be noted that direct comparison between studies should be taken with caution due to differences in types of input data used, sizes of data set and types of models performed. Regarding studies using YOLO models, a previous work [11] reported the sensitivity, specificity, accuracy and precision of a YOLOv3 model in detecting periapical lesions on PR as 92,1%, 76%, 86.3% and 86.4% which were similar to our study. This study trained PAI scores 1 to 5 with 600 periapical root areas for each. Ekert et al. applied CNN with seven layers to detect apical lesions via panoramic radiographs showing the sensitivity and specificity of the model were 95% and 74% [18]. Li et al. used CNN with a dataset of 476 periapical images, 65 of which had lesions. Sensitivity, specificity, accuracy, and precision were 94%, 90%, 92% and 92%, respectively [19]. Recently, a study using panoramic radiography compared YOLOv3 and Faster RCNN. The study showed sensitivity, accuracy, and precision of YOLOv3 model as 91.8%, 77.2%, and 82,9% and of Faster RCNN as 78.8%,74.7%, and 91% respectively [20].

In this current investigation, the performance of Faster R-CNN outperformed YOLOv4 in terms of sensitivity, specificity, accuracy, and precision. The results suggested that Faster R-CNN might be a more effective model for detecting periapical lesions on PR. However, both models have shown promise for automating the detection of periapical lesions. To explain the better performance of Faster R-CNN in our study in detecting periapical lesions, one plausible explanation is that Faster R-CNN is better able to detect subtle differences in the shape of lesions and surrounding tissues, which might be more challenging for YOLOv4. Faster R-CNN uses a region proposal network to identify potential object locations in the image, which is then used to refine the classification and localization of objects [21]. Thus, Faster R-CNN can identify and classify objects even in complex and cluttered images more accurately, for example, in the PR where multiple teeth and anatomical structures are presented in a single image. In contrast, YOLOv4 uses a single-stage object detection approach that directly predicts object bounding boxes and class probabilities based on the entire image [22]. While this approach can be faster and more efficient than the 2-stage approach used by Faster R-CNN, it may also be more prone to errors, especially in cases where the lesions are small or difficult to distinguish from the background such as PAI 3 lesions. We suppose that the two-stage approach of Faster R-CNN was better suited for detecting periapical lesions. Further investigations are needed to confirm our hypothesis.

Another goal of the current study was to evaluate the effects of anatomical factors on the diagnostic performance of the two deep learning models. To achieve this, we divided the upper and lower tooth arches into three separate regions: anterior teeth, premolars, and molars. Interestingly, we did not find clear differences in the performance of both models among these three regions in detecting periapical lesions. Several reasons can be explained for this finding. First, incisors or molars might differ in size and shape but their periapical lesions at the apical roots were similar which can be subject to similar types of lesions. Another possibility is that the consistent quality of the PR may have minimized any variability in the diagnostic accuracy of the models. High visibility and low distortion of PR could contribute to the lack of regional differences in performance. While these explanations are speculative, our finding suggested that the location of the lesion within the tooth arch may not be a major factor affecting the detection of periapical lesions on PR. A previous study also tested the performance of CNN on four types of teeth. The authors found that the CNN performed better on molars than premolars, canines, or incisors with sensitivities of 87%, 63%, 52%, and 64%, respectively [18]. It should be noted that this study was conducted on panoramic radiographs which frequently superimpose ghost images of the spine on anterior teeth [23].

Classifying periapical lesions based on PAI scoring system

The PAI scoring system is a clinically reliable index as it was proven to correlate with histology [24]. However, its potential limitation is the subjective nature of the scoring, which can lead to variability in interpretation among different practitioners. The application of deep learning models as a diagnostic aid could improve reliability and reduce the time and effort required for the diagnosis while remaining comparable accuracy as experienced specialists.

The YOLOv4 showed true prediction in PAI-based classification at 68.06% for PAI 3, 50.94% for PAI 4 and 65.31% for PAI 5 periapical lesions. The true prediction of Faster R-CNN was 89%, 83.01% and 91.84% for PAI 3, PAI 4 and PAI 5 lesions, respectively. The results of the YOLOv4 in our study were an improvement over the previous study by Moidu et al. using YOLOv3 [11], which reported true predictions PAI 3 and 4 at 60% and 71% by YOLOv3. The slightly better performance of YOLOv4 over YOLOv3 could be attributed to the improved architecture and features in YOLOv4. First, YOLOv4 uses a more advanced backbone network called CSPDarknet53, which has deeper layers and a larger receptive field, allowing for better feature extraction and representation. This leads to more accurate object classification. Second, YOLOv4 incorporates various optimization techniques, such as the Mish activation function, SPP block, and PAN path aggregation module, which further improve the model's performance [22].

The true prediction rates achieved by Faster R-CNN were promising for clinical application, as it can potentially aid dentists in accurately classifying periapical lesions. Once again, Faster R-CNN demonstrated superior performance compared to YOLOv4 in classifying periapical lesions. Faster R-CNN is a more complex model that uses a region proposal network to identify potential object locations before making a final classification decision [9]. This may help to reduce false positives and increase the accuracy of the model.

Both models gave the lowest true predictions for PAI 4 which were often misinterpreted as PAI 3, compared to other PAI scores lesions. The misclassification of PAI 4 lesions as PAI 3 can lead to inappropriate treatment decisions and potentially worsen the patient's condition. Therefore, it is important to improve the accuracy of deep learning models in classifying periapical lesions, particularly for PAI 4 lesions. One possible explanation for this misclassification could be the overlap between PAI 3 and PAI 4 lesions in terms of radiographic features. PAI 3 and PAI 4 lesions both exhibit apical periodontitis, but PAI 4 lesions are characterized by a widened apical foramen or a visible radiolucent area beyond the apex [12]. This distinction may not always be clear in radiographs, which could explain why the models struggle to differentiate between the 2 scores. Incorporating additional clinical or radiographic features into the model's training data may aid in the classification of periapical lesions. For example, the presence of a periapical radiolucency with well-defined margins may be more indicative of a PAI 4 lesion, while a poorly defined or diffuse radiolucency may be more indicative of a PAI 3 lesion [25]. The deep learning model can be trained to identify the presence or absence of well-defined margins or diffuse radiolucency in the periapical lesions as part of its decision-making process. The model can learn to assign a higher probability of a lesion being a PAI 4 if it has well-defined margins, and a higher probability of a lesion being a PAI 3 if it has a diffuse radiolucency.

It is worth noting that the two models were trained on a dataset that was biased towards PAI 3 lesions in our study (Table 1). One potential solution to this problem could be to include more data on PAI 4 lesions in the training set. Increasing the diversity of the training dataset by including more cases of PAI 4 lesions in future studies, the models may be better able to learn the features that distinguish PAI 4 from PAI 3 lesions and improve their accuracy in classifying these lesions.

The field of dental applications for deep learning is rapidly evolving beyond classification tasks. A 2023 study by Anum Fatima explored 'multiclass instance segmentation'—a technique that not only detects lesions but also precisely delineated their boundaries [26]. This highlights the ongoing development of diverse deep learning approaches in dentistry. Looking towards the future, the integration of deep learning with dental technology offers exciting possibilities. A 2023 study by Imran Shafi [27] proposed a framework that leverages deep learning for lesion detection within an Internet of Things infrastructure, potentially enabling remote dental health monitoring. While these studies did not directly address PAI classification, they showcased the broader potential of deep learning in dentistry, paving the way for future research that explores its integration with advanced dental technologies.

Limitations

While our study offers valuable insights, several limitations warrant acknowledgment. Firstly, the dataset's composition skewed towards PAI 3 lesions may influence the generalizability of the model's performance across different severity levels. Secondly, our focus on periapical lesions solely detected on periapical radiographs neglects potential complementary information available from other imaging modalities like cone-beam computed tomography (CBCT). Additionally, variations in image quality and patient demographics were not explicitly addressed, potentially impacting the models' performance and generalizability. These limitations underscore the need for cautious interpretation and further investigation.

We recognize the importance of bridging the gap between AI-based diagnostic tools and clinical practice. Future research should incorporate additional clinical insights, including patient demographics, treatment decisions, and long-term follow-up data. By documenting patient characteristics, treatment outcomes, and recurrence rates of periapical lesions, we can provide a more comprehensive evaluation of the diagnostic accuracy of AI models in real-world clinical settings. Furthermore, correlating AI-detected lesions with clinical symptoms and signs of infection can validate the clinical significance of radiographic findings and guide optimal treatment strategies.

Conclusion

Our study demonstrated the potential of YOLOv4 and Faster R-CNN models for detecting and classifying periapical lesions based on the PAI scoring system using PR. The performance of Faster R-CNN outperformed YOLOv4 in both detection and classification.

The location of the lesion within the tooth arch might not be a major factor affecting the detection of periapical lesions on PR.

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Data availability The data that support the findings of this study are available from the corresponding author, upon reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical approval The study was conducted with approval from the Hanoi Medical University Ethics Committee, number HMUIRB463.

References

1. Huumonen S, Suominen A, Vehkalahti M. Prevalence of apical periodontitis in root filled teeth: findings from a nationwide survey in finland. Int Endodontic J. 2017;50(3):229–36.
2. Nair PR. Pathogenesis of apical periodontitis and the causes of endodontic failures. Crit Rev Oral Biol Med. 2004;15(6):348–81.
3. Antony DP, Thomas T, Nivedhitha M. Two-dimensional periapical, panoramic radiography versus three-dimensional cone-beam computed tomography in the detection of periapical lesion after endodontic treatment: a systematic review. Cureus. 2020. https://doi.org/10.7759/cureus.7736.
4. International atomic energy agency, *radiation protection in dental radiology No.108 [IAEA Preprint].* Safety reports series. 2021 **53:**(23) 81.
5. Mazzi-Chaves JF, et al. Cone-beam computed tomographic–based assessment of filled C-shaped canals: artifact expression of cone-beam computed tomography as opposed to micro–computed tomography and nano–computed tomography. J Endodontic. 2020;46(11):1702–11.
6. Corbella S, Srinivas S, Cabitza F. Applications of deep learning in dentistry. Oral Surg Oral Med Oral Pathol Oral Radiol. 2021;132(2):225–38.
7. Arsiwala-Scheppach LT, et al. Machine learning in dentistry: a scoping review. J Clin Med. 2023;12(3):937.
8. Pathak AR, Pandey M, Rautaray S. Application of deep learning for object detection. Procedia computer sci. 2018;132:1706–17.
9. Thanh MTG, et al. Deep learning application in dental caries detection using intraoral photos taken by smartphones. Appl Sci. 2022;12(11):5504.
10. Krois J, et al. Deep learning for the radiographic detection of periodontal bone loss. Sci Rep. 2019;9(1):8495.
11. Moidu NP, et al. Deep learning for categorization of endodontic lesion based on radiographic periapical index scoring system. Clin Oral Investig. 2022;26(1):651–8.
12. Ørstavik D, Kerekes K, Eriksen HM. The periapical index: a scoring system for radiographic assessment of apical periodontitis. Dent Traumatol. 1986;2(1):20–34.
13. Matijević J, et al. Prevalence of apical periodontitis and quality of root canal fillings in population of Zagreb Croatia: a cross-sectional study. Croat Med J. 2011;52(6):679–87.

14. Sidaravicius B, Aleksejuniene J, Eriksen HJDT. Endodontic treatment and prevalence of apical periodontitis in an adult population of Vilnius. Lithuania. 1999;15(5):210–5.
15. Tsuneishi M, et al. Radiographic evaluation of periapical status and prevalence of endodontic treatment in an adult Japanese population. Oral Surg, Oral Med, Oral Pathol, Oral Radiol, Endodontol. 2005;100(5):631–5.
16. Correia-Sousa J, et al. Apical periodontitis and related risk factors: Cross-sectional study. Revis Port de Estomatol, Med Dent{\'a}ria e Cir Maxilofac. 2015;56(4):226–32.
17. Sadr S, et al. Deep learning for detection of periapical radiolucent lesions: a systematic review and meta-analysis of diagnostic test accuracy. J Endod. 2023;49(3):248-261.e3.
18. Ekert T, et al. Deep Learning for the Radiographic Detection of Apical Lesions. J Endod. 2019;45(7):917-922.e5.
19. Li C-W, et al. Detection of dental apical lesions using CNNs on periapical radiograph. Sensors. 2021;21(21):7049.
20. Çelik B, et al. The role of deep learning for periapical lesion detection on panoramic radiographs. Dentomaxillofacial Radiol. 2023. https://doi.org/10.1259/dmfr.20230118.
21. Ren S, et al. Faster r-cnn: Towards real-time object detection with region proposal networks. Adv neural inf process syst. 2015;39:1137–49.
22. Bochkovskiy A, Wang CY, Liao HYM (2020) Yolov4: optimal speed and accuracy of object detection. arXiv preprint. arXiv:2004.10934.
23. Peretz B, Gotler M, Kaffe I. Common errors in digital panoramic radiographs of patients with mixed dentition and patients with permanent dentition. Int J Dent. 2012;2012:584138.
24. Brynolf I. A histological and roentgenological study of the periapical region of human upper incisors, vol. 18. Almqvist & Wiksell; 1967.
25. Zehnder M, Belibasakis GN. A critical analysis of research methods to study clinical molecular biomarkers in endodontic research. Int Endod J. 2022;55:37–45.
26. Fatima A, et al. Deep learning-based multiclass instance segmentation for dental lesion detection. Healthcare. 2023. https://doi.org/10.3390/healthcare11030347.
27. Shafi I, et al. Teeth lesion detection using deep learning and the internet of things post-COVID-19. Sensors. 2023;23(15):6837.